

The Beginner's Guide to Room Config

Your first week with the Room Config Builder

Nothing here needs code, and nothing here needs you to touch a JSON file.

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Read this first

Welcome. If you have been handed this app and a room to set up, this is the short version: the twenty pages that get you from a blank screen to a room that is drawn, priced and deployed.

The full manual is *Room Config Builder - Operation guide*. It answers everything. This one answers the things you hit in the first week, in the order you hit them.

You are never stuck. Press **F1** anywhere in the app and a searchable list of every feature opens over whatever you were doing. Type the word off the button that confused you - "vendor", "spares", "lead time", "who installs it" - and it will find the page for it. Closing it puts you back exactly where you were.

The one idea worth having

You describe the room **once**. Everything else draws itself.

Type in what the room has - a projector, two displays, a switcher, a camera - and the app writes the processor's `config.json` and draws the control schematic, the signal flow, the rack elevation and the cable schedule, and prices the whole thing off your catalog.

So when something looks wrong on a drawing, the fix is almost never on the drawing. It is in the room's description, or in one of the rule files that says how a room turns into a picture. Change it there and every drawing follows.

Step 1: The "one folder" rule

The easiest way to break things is to have files scattered everywhere. Before you do anything else, put all of these in **one folder**:

File or folder	What it is
<code>config.json</code>	The template a new room starts from
<code>ui_schema.json</code>	Tells the app how each setting should look on screen
<code>key_map.json</code>	Translates settings from older config files
<code>buildings.json</code>	Your building names and their codes
<code>processors.json</code>	The rooms you can deploy to, with their IP addresses
<code>av_devices.json</code>	The equipment catalog: connectors, sizes, prices
<code>av_flow_rules.json</code>	How a room turns itself into a drawing
<code>base_costs.json</code>	A typical price per kind of device, for early budgets
<code>labor_rates.json</code>	What an hour of each role costs
<code>devices\</code>	A folder holding the Python control modules
<code>documentation\</code>	PDF manuals, which the app can show you in-app

Anything not there simply isn't used - nothing breaks because a file is missing. But the more of them the folder has, the less you will have to type.

Ask where the shared copy is before you make your own. `av_devices.json`, `ui_schema.json` and `av_flow_rules.json` describe how your shop works, not how your laptop is set up. If your team keeps them on a network drive, point at that - it is how two people drawing the same room draw it the same way.

Step 2: Opening the app for the first time

A setup screen pops up. Don't panic.

1. Click **Browse** next to Root Folder.
2. Select the single folder you made in Step 1.
3. Almost everything else should turn **green (Found)**, because the app looked for them all in that one spot.
4. Click **Finish Setup**.

That's the hard part over. You can change any of it later from the **App Config** tab.

While you are in App Config

Two things there are worth setting now, because they quietly affect every number you will see:

- **Pricing tier.** Every price in the app is published at two: list (MSRP) and the education / institutional price. Pick the one your department buys at.
- **Theme and text size.** Entirely up to you. The app checks its own colors for readability whichever you choose.

Step 3: What you are actually looking at

The tabs run down the left. There are a lot of them, and you will use four in your first week.

Tab	Why you'd go there
Project	The whole building. Which rooms are on the job, what it all costs, who you're buying from.
Cost	What the room you're in comes to.
Wizard	Say what the room <i>has</i> . This is where a new room starts.
Devices	Fill in each device: model, IP address, how it's connected.

The rest - Schematic, AV Flow, Floor Plan, Cabling, Racks - are **drawings the app makes for you**. You don't fill them in; you look at them and correct the room if they're wrong.

Catalog, Schema and **Flow Rules** are settings for the whole app rather than for one room. Leave them alone until something is wrong that they explain.

The rail always fits the window. Make the window small and the labels come off, leaving icons you can hover. You should never have to scroll it.

Step 4: A room, start to finish

Here is the whole loop. It is shorter than it looks.

1. **New Config** - or open an existing room's `config.json`.
2. **Wizard tab**. Set the building, the room number and the room name. Then set how many of each kind of device the room has.
3. **Devices tab**. One sub-tab has appeared per device. Fill each one in: the model, its IP address, how it connects.
4. **System tab**. Which source is plugged into which switcher input, and which display hangs off which output.
5. **Look at the drawings**. Schematic and AV Flow have drawn themselves from what you just typed. If something is missing, it is nearly always a blank number on the System tab.
6. **Cost tab**. The estimate is already there, built from the boxes on the drawing.
7. **Save**, and **Send** to push the config to the processor.

Adding devices - no code required

Never type out code to add a projector or a camera.

- Go to the **Wizard** tab.
- Find the category you want ("Projectors", "Cameras").
- Change the number to how many the room actually has - 0 to 2.
- New tabs appear for those devices, with sensible defaults already filled in.

Take one away the same way. The app removes the block rather than leaving a half-configured device behind in the file.

Step 5: Cheat sheet for the confusing settings

Click a device tab - "Projector 1" - and you get a list of settings. Here is what the tricky ones actually mean.

IP Address / Host

What it is: the network address of the device.

What to do: type the IP address, like `192.168.1.50`. If the device is connected by a **serial cable** straight to the main processor, type `processor1` instead.

Protocol

What it is: the language the device uses to talk over the network.

What to do: pick from the dropdown. Usually **TCP**, **SSH** or **UDP**.

Keep Alive Command

What it is: a "poke" the app sends every so often so the device doesn't fall asleep and drop the connection.

What to do: pick **Power** or **Input**. The app works out which commands are available by reading your Python module files, so the list is already right.

Input

What it is: which plug the device should switch to.

What to do: pick from the list - **HDMI 1**, **HDBaseT**, and so on.

Module

What it is: the Python file that actually drives this device.

What to do: pick the one that matches the model. If nothing in the list claims your model, that's fine - specify it anyway. The app will keep reminding you the driver is missing, which is the point: nobody should find out on site.

Install date

What it is: the day this unit went in.

What to do: fill it in if you know it. It costs you five seconds and it is what the refresh plan is built out of - the year a room comes round again is worked out from this and nothing else.

Step 6: When a drawing looks wrong

The drawings are made from the room, so the fix is in the room. In rough order of likelihood:

What you see	What it usually is
A device is missing from the drawing	Its count on the Wizard is still 0, or its config block was never filled in.
A lead is missing	A blank input or output number on the System tab. Press Draw the routing from config and read the reason it gives.
A box that costs nothing	Its model isn't in the catalog. The Cost tab counts these for you so you know how much of the estimate is guesswork.
A field showing a raw key name in a plain text box	Nothing in <code>ui_schema.json</code> describes it. See the last chapter.
A drawing that no longer matches the room	Press Recreate from config on that tab.

None of these are errors you have caused. A drawing that says something is missing is doing its job.

Step 7: A building, not just a room

Once you have a few rooms, the **Project** tab is where the actual work happens. You don't have to understand all of it on day one, but it helps to know what is there.

- **Rooms** - which configs are on the job, and what each comes to. Rooms nobody has drawn can be typed in as line items.
- **Equipment** - every part on the whole job **once**, quantities added up across rooms. Nine rooms with two transmitters each is eighteen transmitters on one line, which is what a vendor can actually quote.
- **Vendors** - the companies you buy from. Give each one the manufacturers and categories it sells and the parts tag themselves. Use **Pick from the job** rather than typing: the rules match exactly, so "Extron Electronics" against a catalog that says "Extron" quietly claims nothing.
- **Timeline** - when each part has to be *bought*, worked back from the day the building has to be finished.
- **Deliveries** - the purchase orders and what has turned up.
- **Lifecycle** - what the building already has and which year it falls due.
- **Responsibility** - who furnishes and who installs each line of scope. The document that settles "we thought *you* were pulling that conduit".

The room picker is in the title bar, on every tab. Pick a room, or step through them with the arrows, and the editor loads it - the config, the drawing, the racks, the estimate. Sitting on the Cost tab and stepping through eight rooms is a perfectly ordinary thing to do.

Your edits count straight away, but they aren't saved yet. Type a price and the building total moves. The room's *file* hasn't changed until you press **Save room** in the title bar - and the app tells you, in two places, when that is outstanding.

Step 8: Getting the work out

Nearly every screen hands you something, in one of two shapes:

- a **spreadsheet**, for the people who work from it - a contractor prices a total, and a price gets typed into a spreadsheet
- a **picture**, for the people who only have to see it - a submittal, a slide, an email to a dean

The ones you will reach for first:

- **Save All** writes the whole room into one folder: every drawing as a PNG, every report, the workbook, and the config.
- **Workbook** (Project tab) is one spreadsheet with everything about the building.
- **Quote requests** writes one spreadsheet **per vendor**, into a folder you pick. That is the file you email. It has that vendor's parts and nothing else
 - no labor, no tax, no other vendor's pricing.
- **Screenshot & annotate** grabs the tab you're on and lets you draw on it, for the email that starts "the third output is wrong".

Send a vendor the whole workbook instead of their own quote request and you have sent a supplier your margins and a competitor's prices. Use the right button.

Need to change how a setting looks?

If a setting needs a new dropdown option, a better description, or a name your team actually uses, you don't need a programmer.

The friendly way is the **Schema** tab, which is an editor for exactly this. Its **Coverage** list shows every key in your config and whether anything describes it; press **Describe** on one and fill in the label, the help text and what kind of control it should be.

The other way is to open `ui_schema.json` in a text editor, change the text, and press **Reload Schema** on the App Config tab. Both do the same thing.

The same is true for the drawings: if the app puts the wrong receiver in a run or doesn't recognize a kind of source you use, that's a **rule**, and the **Flow Rules** tab is where you change it. It is not a phone call and a wait for the next build.

Where to go next

- Press **F1**. Everything in the app is in there, searchable, and it is always current.
- Read *Room Config Builder - Operation guide* for the full version of anything here.
- The chapters in that guide worth reading early are **A room, start to finish**, **The job**, and **When something looks wrong**.